

SOLIDWORKS API (.NET VBA C#) — Reference (152/337): IMassProperty

SOLIDWORKS API Documentation (converted from CHM)

IMassProperty (Type)

IMassProperty Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMassProperty Interface

Obsolete. Superseded by [IMassProperty2](#).

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMassProperty
```

Visual Basic (Usage)

```
Dim instance As IMassProperty
```

C#

```
public interface IMassProperty
```

C++/CLI

```
public interface class IMassProperty
```

Visual Basic for Applications (VBA) Syntax

See [MassProperty](#).

Example

[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

Remarks

See [Mass Properties](#) for more information.

Accessors

[ModelDocExtension::CreateMassProperty](#)

See Also

[IMassProperty Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDocExtension::GetMassProperties2 Method \(\)](#)

IMassProperty Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMassProperty Interface

The following tables list the members exposed by [IMassProperty](#).

Public Properties

Name	Description
CenterOfMass	Gets the center of mass.
Density	Gets the density for this model.

 Mass	Gets the mass of this model.
 OverrideCenterOfMass	Gets or sets whether to override the calculated center of mass value for the model.
 OverrideMass	Gets or sets whether to override the calculated mass value for the model.
 OverrideMomentsOfInertia	Gets or sets whether to override the calculated moments of inertia for the model.
 PrincipalAxesOfInertia	Gets the principal axes of inertia for this model.
 PrincipalMomentsOfInertia	Gets the principal moments of inertia for this model.
 SurfaceArea	Gets the surface area for this model.
 UserAssigned	Gets whether the mass property values are user-defined or calculated for this document, regardless of which model is being edited.
 UseSystemUnits	Gets and sets whether to use system units or document units when calculating mass properties for this model.
 Volume	Gets the volume of this model.

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Public Methods

Name	Description
 AddBodies	Uses only the specified bodies when calculating the mass properties for this model.
 GetMomentOfInertia	Gets the moment of inertia at the specified coordinate system for this model.
 IAddBodies	Uses only the specified bodies when calculating the mass properties for this model.
 IGetCenterOfMass	Gets the center of mass for this model.
 IGetMomentOfInertia	Gets the moment of inertia at the specified coordinate system for this model.
 IGetPrincipalAxesOfInertia	Gets the principal axes of inertia for this model.
 IGetPrincipalMomentsOfInertia	Gets the principal moments of inertia for this model.
 ISetAssignedMassProp	Sets the mass and center of gravity for the specified configurations for this model being edited in this part or assembly document.
 ISetOverrideCenterOfMassValue	Overrides the center of mass of the model currently being edited in this part or assembly document.
 ISetOverrideMassValue	Overrides the mass of the model currently being edited in this part or assembly document.
 ISetOverrideMomentsOfInertiaValue	Overrides the moments of inertia of the model currently being edited in this part or assembly document.
 ISetOverridePrincipalAxesOrientation	Overrides the orientation of the specified principal axis of inertia for the model currently being edited in this part or assembly document.
 ISetOverridePrincipalMomentsOfInertia	Overrides the principal moments of inertia of the model currently being edited in this part or assembly document.
 SetAssignedMassProp	Sets the mass and center of gravity for the specified configurations for this model being edited in this part or assembly document.
 SetCoordinateSystem	Sets the coordinate system to use when calculating mass properties for this model.
 SetOverrideCenterOfMassValue	Overrides the center of mass of the model currently being edited in this part or assembly document.
 SetOverrideMassValue	Overrides the mass of the model currently being edited in this part or assembly document.
 SetOverrideMomentsOfInertiaValue	Overrides the moments of inertia of the model currently being edited in this part or assembly document.
 SetOverridePrincipalAxesOrientation	Overrides the orientation of the specified principal axis of inertia for the model currently being edited in this part or assembly document.
 SetOverridePrincipalMomentsOfInertia	Overrides the principal moments of inertia of the model currently being edited in this part or assembly document.

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See Also

[IMassProperty Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDocExtension::GetMassProperties2 Method \(\)](#)

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IMassProperty Interface Methods

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IMassProperty Interface

For a list of all members of this type, see [IMassProperty members](#).

Public Methods

Name	Description
 AddBodies	Uses only the specified bodies when calculating the mass properties for this model.
 GetMomentOfInertia	Gets the moment of inertia at the specified coordinate system for this model.
 IAddBodies	Uses only the specified bodies when calculating the mass properties for this model.
 IGetCenterOfMass	Gets the center of mass for this model.
 IGetMomentOfInertia	Gets the moment of inertia at the specified coordinate system for this model.
 IGetPrincipleAxesOfInertia	Gets the principal axes of inertia for this model.
 IGetPrincipleMomentsOfInertia	Gets the principal moments of inertia for this model.
 ISetAssignedMassProp	Sets the mass and center of gravity for the specified configurations for this model being edited in this part or assembly document.
 ISetOverrideCenterOfMassValue	Overrides the center of mass of the model currently being edited in this part or assembly document.
 ISetOverrideMassValue	Overrides the mass of the model currently being edited in this part or assembly document.
 ISetOverrideMomentsOfInertiaValue	Overrides the moments of inertia of the model currently being edited in this part or assembly document.
 ISetOverridePrincipleAxesOrientation	Overrides the orientation of the specified principal axis of inertia for the model currently being edited in this part or assembly document.
 ISetOverridePrincipleMomentsOfInertia	Overrides the principal moments of inertia of the model currently being edited in this part or assembly document.
 SetAssignedMassProp	Sets the mass and center of gravity for the specified configurations for this model being edited in this part or assembly document.
 SetCoordinateSystem	Sets the coordinate system to use when calculating mass properties for this model.
 SetOverrideCenterOfMassValue	Overrides the center of mass of the model currently being edited in this part or assembly document.
 SetOverrideMassValue	Overrides the mass of the model currently being edited in this part or assembly document.
 SetOverrideMomentsOfInertiaValue	Overrides the moments of inertia of the model currently being edited in this part or assembly document.
 SetOverridePrincipleAxesOrientation	Overrides the orientation of the specified principal axis of inertia for the model currently being edited in this part or assembly document.
 SetOverridePrincipleMomentsOfInertia	Overrides the principal moments of inertia of the model currently being edited in this part or assembly document.

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See Also

[IMassProperty Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDocExtension::GetMassProperties2 Method \(\)](#)

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IMassProperty Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMassProperty Interface

For a list of all members of this type, see [IMassProperty members](#).

Public Properties

Name	Description
 CenterOfMass	Gets the center of mass.
 Density	Gets the density for this model.
 Mass	Gets the mass of this model.
 OverrideCenterOfMass	Gets or sets whether to override the calculated center of mass value for the model.
 OverrideMass	Gets or sets whether to override the calculated mass value for the model.
 OverrideMomentsOfInertia	Gets or sets whether to override the calculated moments of inertia for the model.
 PrincipleAxesOfInertia	Gets the principal axes of inertia for this model.
 PrincipleMomentsOfInertia	Gets the principal moments of inertia for this model.
 SurfaceArea	Gets the surface area for this model.

 UserAssigned	Gets whether the mass property values are user-defined or calculated for this document, regardless of which model is being edited.
 UseSystemUnits	Gets and sets whether to use system units or document units when calculating mass properties for this model.
 Volume	Gets the volume of this model.

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See Also

[IMassProperty Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDocExtension::GetMassProperties2 Method \(\)](#)

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AddBodies Method (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : AddBodies Method (IMassProperty)

Uses only the specified bodies when calculating the mass properties for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function AddBodies( _
    ByVal BodyListIn As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim BodyListIn As System.Object
Dim value As System.Boolean

value = instance.AddBodies(BodyListIn)
```

C#

```
System.bool AddBodies(
    System.object BodyListIn
)
```

C++/CLI

```
System.bool AddBodies(
    & System.Object^ BodyListIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyListIn
Array of [bodies](#) in the model

Return Value

True if the specified bodies are used for calculating mass properties, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty::AddBodies](#).

Example

[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

Remarks

This method lets you select which solid bodies to use when calculating mass properties, much like a end-user selecting the bodies via the user interface.

NOTE: If a sheet (surface) body is specified, it is ignored and false is returned.

The results of the mass property calculations vary based on whether or not this method is used.

If this method is...	Then...
Called and bodies are specified	These bodies can either be from a subset of the documents body list or from temporary bodies. NOTE: Each specified body should either come from the owning document or be a temporary body. If the body does not satisfy either case, then it is not used when calculating the mass properties.
Not called	Mass properties' calculations include all available bodies in the document. Part. All of the solid bodies are included in the calculations. Assembly. All of the bodies in all of the components are used in the calculations.

If the document from which the MassProperty object came is an assembly, then any body from any of the child components can be used. To obtain the body, call [IComponent2::GetBodies2](#). Your application does not need to make a copy of the body or apply a transform to the body.

 See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::IAddBodies Method](#)

 Availability

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CenterOfMass Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : CenterOfMass Property (IMassProperty)

Gets the center of mass.

 .NET Syntax

Visual Basic (Declaration)

ReadOnly Property CenterOfMass As System.Object

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim value As System.Object

value = instance.CenterOfMass
```

C#

```
System.object CenterOfMass {get;}
```

C++/CLI

```
property System.Object^ CenterOfMass {
    System.Object^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array containing the center of mass coordinates (see **Remarks**)

 Visual Basic for Applications (VBA) Syntax

See [MassProperty::CenterOfMass](#).

Example

[Insert Coordinate System at Center of Mass \(VBA\)](#)
[Set Dimensions to Mid-tolerance \(VBA\)](#)
[Get and Override Mass Properties \(C#\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(VBA\)](#)
[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

Remarks

This property returns metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles:

[x, y, z].

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::IGetCenterOfMass Method](#)
[IMassProperty::ISetAssignedMassProp Method](#)
[IMassProperty::SetAssignedMassProp Method](#)
[IMassProperty::Mass Property](#)
[IMassProperty::OverrideCenterOfMass Property](#)

Availability

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Density Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : Density Property (IMassProperty)

Gets the density for this model.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Density As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim value As System.Double

value = instance.Density
```

C#

```
System.double Density {get;}
```

C++/CLI

```
property System.double Density {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Density

Visual Basic for Applications (VBA) Syntax

See [MassProperty::Density](#).

Example

[Set Dimensions to Mid-tolerance \(VBA\)](#)
[Get and Override Mass Properties \(C#\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(VBA\)](#)
[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

Remarks

You can also:

- Derive the density of the body by calculating $Density = (Mass / Volume)$.
- Use [ISldWorks::GetUserPreferenceDoubleValue](#) to get the density of the model.

This property returns metric units unless explicitly specified otherwise.

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)

Availability

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GetMomentOfInertia Method (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : GetMomentOfInertia Method (IMassProperty)

Gets the moment of inertia at the specified coordinate system for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMomentOfInertia( _  
    ByVal WhereTaken As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim WhereTaken As System.Integer  
Dim value As System.Object  
  
value = instance.GetMomentOfInertia(WhereTaken)
```

C#

```
System.object GetMomentOfInertia(  
    System.int WhereTaken  
)
```

C++/CLI

```
System.Object^ GetMomentOfInertia(  
&    System.int WhereTaken  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhereTaken

Coordinate system as defined in [swMassPropertyMoment_e](#)

Return Value

Array containing the moment of inertia calculations (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [MassProperty::GetMomentOfInertia](#).

Example

[Get and Override Mass Properties \(VBA\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

Remarks

This method returns metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles:

[*Lxx*, *Lxy*, *Lxz*, *Lyx*, *Lyy*, *Lyz*, *Lzx*, *Lzy*, *Lzz*]

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::IGetMomentOfInertia Method](#)
[IMassProperty::IGetPrincipleAxesOfInertia Method](#)
[IMassProperty::IGetPrincipleMomentsOfInertia Method](#)
[IMassProperty::PrincipleAxesOfInertia Property](#)
[IMassProperty::PrincipleMomentsOfInertia Property](#)

Availability

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IAddBodies Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : IAddBodies Method (IMassProperty)

Uses only the specified bodies when calculating the mass properties for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function IAddBodies( _  
    ByVal NumBodies As System.Integer, _  
    ByRef BodyListIn As Body2 _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim NumBodies As System.Integer  
Dim BodyListIn As Body2  
Dim value As System.Boolean
```

```
value = instance.IAddBodies(NumBodies, BodyListIn)
```

C#

```
System.bool IAddBodies(  
    System.int NumBodies,  
    ref Body2 BodyListIn  
)
```

C++/CLI

```
System.bool IAddBodies(
& System.int NumBodies,
& Body2^% BodyListIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumBodies

Number of bodies in the model

BodyListIn

- in-process, unmanaged C++: Pointer to an array of [bodies](#) in the model of size NumBodies

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if the specified bodies are used for calculating mass properties, false if not

Remarks

This method lets you select which solid bodies to use when calculating mass properties, much like a end-user selecting the bodies via the interface.

NOTE: If a sheet (surface) body is specified, it is ignored and false is returned.

The results of the mass property calculations vary based on whether or not this method is used.

If this method is...	Then...
Called and bodies are specified	These bodies can either be from a subset of the documents body list or from temporary bodies. NOTE: Each specified body should either come from the owning document or be a temporary body. If the body does not satisfy either case, then it is not used when calculating the mass properties.
Not called	Mass properties' calculations include all available bodies in the document. • Part. All of the solid bodies are included in the calculations. • Assembly. All of the bodies in all of the components are used in the calculations.

If the document from which the MassProperty object came is an assembly, then any body from any of the child components can be used. To obtain the body, call [IComponent2::GetBodies2](#). Your application does not need to make a copy of the body or apply a transform to the body.

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::AddBodies Method](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetCenterOfMass Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : IGetCenterOfMass Method (IMassProperty)

Gets the center of mass for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetCenterOfMass() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim value As System.Double

value = instance.IGetCenterOfMass()
```

C#

```
System.double IGetCenterOfMass()
```

C++/CLI

```
System.double IGetCenterOfMass();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of size 3 (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

This method returns metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles as follows:

[X, Y, Z]

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty.CenterOfMass Property](#)

Availability

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IGetMomentOfInertia Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : IGetMomentOfInertia Method (IMassProperty)

Gets the moment of inertia at the specified coordinate system for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMomentOfInertia( _
    ByVal WhereTaken As System.Integer _
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IMassProperty
Dim WhereTaken As System.Integer
Dim value As System.Double
```

```
value = instance.IGetMomentOfInertia(WhereTaken)
```

C#

```
System.double IGetMomentOfInertia(
    System.int WhereTaken
)
```

C++/CLI



```
System.double IGetMomentOfInertia(  
& System.int WhereTaken  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhereTaken

Coordinate system as defined in [swMassPropertyMoment_e](#)

Return Value

- in-process, unmanaged C++: Pointer to an array of size 9 containing the moment of inertia calculations (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method returns metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles:

[*Lxx*, *Lxy*, *Lxz*, *Lyx*, *Lyy*, *Lyz*, *Lzx*, *Lzy*, *Lzz*]

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::IGetMomentOfInertia Method](#)
[IMassProperty::IGetPrincipleAxesOfInertia Method](#)
[IMassProperty::IGetPrincipleMomentsOfInertia Method](#)
[IMassProperty::PrincipleAxesOfInertia Property](#)
[IMassProperty::PrincipleMomentsOfInertia Property](#)

Availability

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IGetPrincipleAxesOfInertia Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : IGetPrincipleAxesOfInertia Method (IMassProperty)

Gets the principal axes of inertia for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPrincipleAxesOfInertia( _  
    ByVal Axis As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim Axis As System.Integer  
Dim value As System.Double
```

```
value = instance.IGetPrincipleAxesOfInertia(Axis)
```

C#

```
System.double IGetPrincipleAxesOfInertia(  
    System.int Axis  
)
```

C++/CLI

```
System.double IGetPrincipleAxesOfInertia(  
& System.int Axis
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Axis

- 0 = X axis
- 1 = Y axis
- 2 = Z axis

Return Value

- in-process, unmanaged C++: Pointer to an array of size 3 (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method returns:

- a vector that represents the requested axis.
- metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles:

[*lx*, *ly*, *lz*]

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::GetMomentOfInertia Method](#)
[IMassProperty::IGetMomentOfInertia Method](#)
[IMassProperty::IGetPrincipleMomentsOfInertia Method](#)
[IMassProperty::PrincipleAxesOfInertia Property](#)
[IMassProperty::PrincipleMomentsOfInertia Property](#)

Availability

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IGetPrincipleMomentsOfInertia Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : IGetPrincipleMomentsOfInertia Method (IMassProperty)

Gets the principal moments of inertia for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPrincipleMomentsOfInertia() As System.Double
```

Visual Basic (Usage)

```
Dim instance As IMassProperty  
Dim value As System.Double  
  
value = instance.IGetPrincipleMomentsOfInertia()
```

C#

```
System.double IGetPrincipleMomentsOfInertia()
```

C++/CLI

```
System.double IGetPrincipleMomentsOfInertia();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of size 3 (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

This method returns metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles:

[*Px*, *Py*, *Pz*]

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::GetMomentOfInertia Method](#)
[IMassProperty::IGetMomentOfInertia Method](#)
[IMassProperty::IGetPrincipleAxesOfInertia Method](#)
[IMassProperty::PrincipleAxesOfInertia Property](#)
[IMassProperty::PrincipleMomentsOfInertia Property](#)

Availability

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ISetAssignedMassProp Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : ISetAssignedMassProp Method (IMassProperty)

Sets the mass and center of gravity for the specified configurations for this model being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetAssignedMassProp( _  
    ByVal Mass As System.Double, _  
    ByVal Center\_x As System.Double, _  
    ByVal Center\_y As System.Double, _  
    ByVal Center\_z As System.Double, _  
    ByVal Config\_opt As System.Integer, _  
    ByVal ConfigNum As System.Integer, _  
    ByRef Config\_names As System.String _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim Mass As System.Double  
Dim Center_x As System.Double  
Dim Center_y As System.Double  
Dim Center_z As System.Double  
Dim Config_opt As System.Integer  
Dim ConfigNum As System.Integer  
Dim Config_names As System.String  
Dim value As System.Boolean
```

```
value = instance.ISetAssignedMassProp(Mass, Center_x, Center_y, Center_z, Config_opt, ConfigNum,  
    Config_names)
```

C#

```
System.bool ISetAssignedMassProp(  
    System.double Mass,  
    System.double Center\_x,  
    System.double Center\_y,  
    System.double Center\_z,  
    System.int Config\_opt,  
    System.int ConfigNum,
```

```
ref System.string Config\_names
)
```

C++/CLI

```
System.bool ISetAssignedMassProp(
& System.double Mass,
& System.double Center\_x,
& System.double Center\_y,
& System.double Center\_z,
& System.int Config\_opt,
& System.int ConfigNum,
& System.String^% Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Mass
Value for mass
Center_x
x coordinate for center of gravity
Center_y
y coordinate for center of gravity
Center_z
z coordinate for center of gravity

Config_opt
Configuration options as defined in [swInConfigurationOpts_e](#)

ConfigNum
Number of configurations
Config_names

- in-process, unmanaged C++: Pointer to an array of configuration names of size ConfigNum
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if the mass and center of gravity are set, false if not

 Remarks

If...	Then...
You are editing a subcomponent	you are setting the mass for this subcomponent and not for the top-level model.
You specify a value < 0 for mass	mass is calculated; it is not user-defined.
Config_opt is set to swSpecifyConfiguration	Config_names is used.

 See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::SetAssignedMassProp Method](#)
[IMassProperty::IGetCenterOfMass Method](#)
[IMassProperty::CenterOfMass Property](#)
[IMassProperty::Mass Property](#)
[IMassProperty::UserAssigned Property](#)

 Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetOverrideCenterOfMassValue Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : ISetOverrideCenterOfMassValue Method (IMassProperty)

Overrides the center of mass of the model currently being edited in this part or assembly document.

 .NET Syntax

Visual Basic (Declaration)

```

Function ISetOverrideCenterOfMassValue( _
    ByRef Value As System.Double, _
    ByVal CoordinateSystemName As System.String, _
    ByVal Config\_option As System.Integer, _
    ByVal Config\_numbers As System.Integer, _
    ByRef Config\_names As System.String _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IMassProperty
Dim Value As System.Double
Dim CoordinateSystemName As System.String
Dim Config_option As System.Integer
Dim Config_numbers As System.Integer
Dim Config_names As System.String
Dim value As System.Boolean

value = instance.ISetOverrideCenterOfMassValue(Value, CoordinateSystemName, Config_option,
Config_numbers, Config_names)

```

C#

```

System.bool ISetOverrideCenterOfMassValue(
    ref System.double Value,
    System.string CoordinateSystemName,
    System.int Config\_option,
    System.int Config\_numbers,
    ref System.string Config\_names
)

```

C++/CLI

```

System.bool ISetOverrideCenterOfMassValue(
    & System.double% Value,
    & System.String^ CoordinateSystemName,
    & System.int Config\_option,
    & System.int Config\_numbers,
    & System.String^% Config\_names
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value

- in-process, unmanaged C++: Pointer to an array of three doubles of the x, y, and z coordinates of the center of mass
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

CoordinateSystemName

Name of the coordinate system in which the center of mass is defined

Config_option

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_numbers

Number of configurations

Config_names

- in-process, unmanaged C++: Pointer to an array of configuration names of size Config_numbers; valid only if Config_option = swInConfigurationOpts_e.swSpecifyConfiguration
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if the center of mass is overridden, false if not

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::SetOverrideCenterOfMassValue Method](#)
[IMassProperty::OverrideCenterOfMass Property](#)

Availability

ISetOverrideMassValue Method (IMassProperty)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : ISetOverrideMassValue Method (IMassProperty)

Overrides the mass of the model currently being edited in this part or assembly document.

 .NET Syntax

Visual Basic (Declaration)

```
Function ISetOverrideMassValue( _
    ByVal Value As System.Double, _
    ByVal Config\_option As System.Integer, _
    ByVal Config\_numbers As System.Integer, _
    ByRef Config\_names As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim Value As System.Double
Dim Config_option As System.Integer
Dim Config_numbers As System.Integer
Dim Config_names As System.String
Dim value As System.Boolean

value = instance.ISetOverrideMassValue(Value, Config_option, Config_numbers, Config_names)
```

C#

```
System.bool ISetOverrideMassValue(
    System.double Value,
    System.int Config\_option,
    System.int Config\_numbers,
    ref System.string Config\_names
)
```

C++/CLI

```
System.bool ISetOverrideMassValue(
    & System.double Value,
    & System.int Config\_option,
    & System.int Config\_numbers,
    & System.String^% Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

*Value*Override mass value (see **Remarks**)*Config_option*Configuration option as defined in [swlnConfigurationOptions](#)*Config_numbers*

Number of configurations

Config_names

- In-process, unmanaged C++: Pointer to an array of configuration names of size Config_numbers (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if the mass value is overridden, false if not

 Remarks

If...

Then...

You are editing a subcomponent

Value > 0

Value < 0

Config_option =
[swlnConfigurationOpts_e](#).swSpecifyConfiguration

you are overriding the mass for this subcomponent and not for the top-level model.

mass is overridden.

mass is calculated.

Config_names is used.

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::SetOverrideMassValue Method](#)
[IMassProperty::OverrideMass Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetOverrideMomentsOfInertiaValue Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : ISetOverrideMomentsOfInertiaValue Method (IMassProperty)

Overrides the moments of inertia of the model currently being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetOverrideMomentsOfInertiaValue( _  
    ByVal ReferenceFrame As System.Integer, _  
    ByVal CoordinateSystemName As System.String, _  
    ByRef Value As System.Double, _  
    ByVal Config\_option As System.Integer, _  
    ByVal Config\_numbers As System.Integer, _  
    ByRef Config\_names As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim ReferenceFrame As System.Integer  
Dim CoordinateSystemName As System.String  
Dim Value As System.Double  
Dim Config_option As System.Integer  
Dim Config_numbers As System.Integer  
Dim Config_names As System.String  
Dim value As System.Boolean
```

```
value = instance.ISetOverrideMomentsOfInertiaValue(ReferenceFrame, CoordinateSystemName, Value,  
Config_option, Config_numbers, Config_names)
```

C#

```
System.bool ISetOverrideMomentsOfInertiaValue(  
    System.int ReferenceFrame,  
    System.string CoordinateSystemName,  
    ref System.double Value,  
    System.int Config\_option,  
    System.int Config\_numbers,  
    ref System.string Config\_names  
)
```

C++/CLI

```

System.bool ISetOverrideMomentsOfInertiaValue(
& System.int ReferenceFrame,
& System.String^ CoordinateSystemName,
& System.double% Value,
& System.int Config\_option,
& System.int Config\_numbers,
& System.String^% Config\_names
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ReferenceFrame

Frame of reference as defined in [swMomentsOfInertiaReferenceFrame_e](#)

CoordinateSystemName

Name of coordinate system; valid only if ReferenceFrame = swMomentsOfInertiaReferenceFrame_e.swMomentsOfInertiaReferenceFrame_UserCoordinateSystem

- in-process, unmanaged C++: Pointer to an array of nine doubles: [Lxx, Lxy, Lxz, Lyx, Lyy, Lyz, Lzx, Lzy, Lzz]
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Config_option

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_numbers

Number of configurations

Config_names

- in-process, unmanaged C++: Pointer to an array of configuration names of size Config_numbers; valid only if Config_option = swInConfigurationOpts_e.swSpecifyConfiguration
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if the moments of inertia are overridden, false if not

See Also

[IMassProperty Interface](#)

[IMassProperty Members](#)

[IMassProperty::SetOverrideMomentsOfInertiaValue Method](#)

[IMassProperty::OverrideMomentsOfInertia Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetOverridePrincipleAxesOrientation Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : ISetOverridePrincipleAxesOrientation Method (IMassProperty)

Overrides the orientation of the specified principal axis of inertia for the model currently being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```

Function ISetOverridePrincipleAxesOrientation( _
    ByVal Axis As System.Integer, _
    ByRef Value As System.Double, _
    ByVal Config\_option As System.Integer, _
    ByVal Config\_numbers As System.Integer, _
    ByRef Config\_names As System.String _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IMassProperty
Dim Axis As System.Integer
Dim Value As System.Double
Dim Config_option As System.Integer
Dim Config_numbers As System.Integer
Dim Config_names As System.String
Dim value As System.Boolean

```

```
value = instance.ISetOverridePrincipleAxesOrientation(Axis, Value, Config_option, Config_numbers, Config_names)
```

C#

```
System.bool ISetOverridePrincipleAxesOrientation(  
    System.int Axis,  
    ref System.double Value,  
    System.int Config\_option,  
    System.int Config\_numbers,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.bool ISetOverridePrincipleAxesOrientation(  
& System.int Axis,  
& System.double% Value,  
& System.int Config\_option,  
& System.int Config\_numbers,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Axis

One of the following principal axes:

- 0 = X axis
- 1 = Y axis
- 2 = Z axis

Value

- in-process, unmanaged C++: Pointer to an array of three doubles of the x, y, and z coordinates of the specified principal Axis
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Config_option

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_numbers

Number of configurations

Config_names

- in-process, unmanaged C++: Pointer to an array of configuration names of size Config_numbers; valid only if Config_option = swInConfigurationOpts_e.swSpecifyConfiguration
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if orientation of principal axis is overridden, false if not

See Also

[IMassProperty Interface](#)

[IMassProperty Members](#)

[IMassProperty::SetOverridePrincipleAxesOrientation Method](#)

[IMassProperty::PrincipleAxesOfInertia Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetOverridePrincipleMomentsOfInertia Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : ISetOverridePrincipleMomentsOfInertia Method (IMassProperty)

Overrides the principal moments of inertia of the model currently being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetOverridePrincipleMomentsOfInertia( _  
    ByRef Value As System.Double, _  
    ByVal Config\_option As System.Integer, _  
    ByVal Config\_numbers As System.Integer, _
```

```
ByRef Config\_names As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim Value As System.Double  
Dim Config_option As System.Integer  
Dim Config_numbers As System.Integer  
Dim Config_names As System.String  
Dim value As System.Boolean  
  
value = instance.ISetOverridePrincipleMomentsOfInertia(Value, Config_option, Config_numbers,  
Config_names)
```

C#

```
System.bool ISetOverridePrincipleMomentsOfInertia(  
    ref System.double Value,  
    System.int Config\_option,  
    System.int Config\_numbers,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.bool ISetOverridePrincipleMomentsOfInertia(  
& System.double% Value,  
& System.int Config\_option,  
& System.int Config\_numbers,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value

- in-process, unmanaged C++: Pointer to an array of three doubles of the principal moments of inertia: [Px, Py, Pz]
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Config_option

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_numbers

Number of configurations

Config_names

- in-process, unmanaged C++: Pointer to an array of configuration names of size Config_numbers; valid only if Config_option = swInConfigurationOpts_e.swSpecifyConfiguration
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

See Also

[IMassProperty Interface](#)

[IMassProperty Members](#)

[IMassProperty::SetOverridePrincipleMomentsOfInertia Method](#)

[IMassProperty::OverrideMomentsOfInertia Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Mass Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : Mass Property (IMassProperty)

Gets the mass of this model.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Mass As System.Double

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim value As System.Double
```

```
value = instance.Mass
```

C#

```
System.double Mass {get;}
```

C++/CLI

```
property System.double Mass {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Mass

Visual Basic for Applications (VBA) Syntax

See [MassProperty::Mass](#).

Example

[Insert Coordinate System at Center of Mass \(VBA\)](#)
[Set Dimensions to Mid-tolerance \(VBA\)](#)
[Get and Override Mass Properties \(C#\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(VBA\)](#)
[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::OverrideMass Property](#)

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

OverrideCenterOfMass Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : OverrideCenterOfMass Property (IMassProperty)

Gets or sets whether to override the calculated center of mass value for the model.

.NET Syntax

Visual Basic (Declaration)

```
Property OverrideCenterOfMass As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim value As System.Boolean
```

```
instance.OverrideCenterOfMass = value
```

```
value = instance.OverrideCenterOfMass
```

C#

```
System.bool OverrideCenterOfMass {get; set;}
```

C++/CLI

```
property System.bool OverrideCenterOfMass {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to override the center of mass value, false to use the calculated value

Visual Basic for Applications (VBA) Syntax

See [MassProperty::OverrideCenterOfMass](#).

Example

[Get and Override Mass Properties \(VBA\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(C#\)](#)

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::ISetOverrideCenterOfMassValue Method](#)
[IMassProperty::SetOverrideCenterOfMassValue Method](#)
[IMassProperty::CenterOfMass Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

OverrideMass Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : OverrideMass Property (IMassProperty)

Gets or sets whether to override the calculated mass value for the model.

.NET Syntax

Visual Basic (Declaration)

```
Property OverrideMass As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim value As System.Boolean
```

```
instance.OverrideMass = value
```

```
value = instance.OverrideMass
```

C#

```
System.bool OverrideMass {get; set;}
```

C++/CLI

```
property System.bool OverrideMass {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to override the mass value, false to use the calculated value

Visual Basic for Applications (VBA) Syntax

See [MassProperty::OverrideMass](#).

Example

[Get and Override Mass Properties \(VBA\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(C#\)](#)

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::ISetOverrideMassValue Method](#)
[IMassProperty::SetOverrideMassValue Method](#)
[IMassProperty::Mass Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

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OverrideMomentsOfInertia Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : OverrideMomentsOfInertia Property (IMassProperty)

Gets or sets whether to override the calculated moments of inertia for the model.

.NET Syntax

Visual Basic (Declaration)

```
Property OverrideMomentsOfInertia As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim value As System.Boolean

instance.OverrideMomentsOfInertia = value

value = instance.OverrideMomentsOfInertia
```

C#

```
System.bool OverrideMomentsOfInertia {get; set;}
```

C++/CLI

```
property System.bool OverrideMomentsOfInertia {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to override the moments of inertia, false to use the calculated values

Visual Basic for Applications (VBA) Syntax

See [MassProperty::OverrideMomentsOfInertia](#).

Example

[Get and Override Mass Properties \(VBA\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(C#\)](#)

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::ISetOverrideMomentsOfInertiaValue Method](#)
[IMassProperty::SetOverrideMomentsOfInertiaValue Method](#)
[IMassProperty::PrincipalMomentsOfInertia Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

PrincipleAxesOfInertia Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : PrincipleAxesOfInertia Property (IMassProperty)

Gets the principal axes of inertia for this model.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property PrincipleAxesOfInertia( _  
    ByVal Axis As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim Axis As System.Integer  
Dim value As System.Object  
  
value = instance.PrincipleAxesOfInertia(Axis)
```

C#

```
System.object PrincipleAxesOfInertia(  
    System.int Axis  
) {get;}
```

C++/CLI

```
property System.Object^ PrincipleAxesOfInertia {  
    System.Object^ get(System.int Axis);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Axis

- 0 = x axis
- 1 = y axis
- 2 = z axis

Property Value

Array of size 3 (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [MassProperty.PrincipleAxesOfInertia](#).

Example

[Insert Coordinate System at Center of Mass \(VBA\)](#)
[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

Remarks

This method returns:

- a vector that represents the requested axis.
- metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles:

[*lx*, *ly*, *lz*]

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::GetMomentOfInertia Method](#)
[IMassProperty::IGetMomentOfInertia Method](#)
[IMassProperty::IGetPrincipleMomentsOfInertia Method](#)
[IMassProperty::PrincipleMomentsOfInertia Property](#)
[IMassProperty::IGetPrincipleAxesOfInertia Method](#)
[IMassProperty::ISetOverridePrincipleAxesOrientation Method](#)
[IMassProperty::SetOverridePrincipleAxesOrientation Method](#)

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

PrincipleMomentsOfInertia Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : PrincipleMomentsOfInertia Property (IMassProperty)

Gets the principal moments of inertia for this model.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property PrincipleMomentsOfInertia As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim value As System.Object

value = instance.PrincipleMomentsOfInertia
```

C#

```
System.object PrincipleMomentsOfInertia {get;}
```

C++/CLI

```
property System.Object^ PrincipleMomentsOfInertia {
    System.Object^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of size 3 (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [MassProperty.PrincipleMomentsOfInertia](#).

Example

[Get and Override Mass Properties \(C#\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(VBA\)](#)
[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

Remarks

This property returns metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles:

[*Px*, *Py*, *Pz*]

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::GetMomentOfInertia Method](#)
[IMassProperty::IGetMomentOfInertia Method](#)
[IMassProperty::IGetPrincipleAxesOfInertia Method](#)
[IMassProperty::IGetPrincipleMomentsOfInertia Method](#)
[IMassProperty.PrincipleAxesOfInertia Property](#)
[IMassProperty.OverrideMomentsOfInertia Property](#)

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetAssignedMassProp Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : SetAssignedMassProp Method (IMassProperty)

Sets the mass and center of gravity for the specified configurations for this model being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function SetAssignedMassProp( _  
    ByVal Mass As System.Double, _  
    ByVal Center\_x As System.Double, _  
    ByVal Center\_y As System.Double, _  
    ByVal Center\_z As System.Double, _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_names As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim Mass As System.Double  
Dim Center_x As System.Double  
Dim Center_y As System.Double  
Dim Center_z As System.Double  
Dim Config_opt As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Boolean
```

```
value = instance.SetAssignedMassProp(Mass, Center_x, Center_y, Center_z, Config_opt, Config_names)
```

C#

```
System.bool SetAssignedMassProp(  
    System.double Mass,
```

```
System.double Center\_x,
System.double Center\_y,
System.double Center\_z,
System.int Config\_opt,
System.object Config\_names
)
```

C++/CLI

```
System.bool SetAssignedMassProp(
& System.double Mass,
& System.double Center\_x,
& System.double Center\_y,
& System.double Center\_z,
& System.int Config\_opt,
& System.Object^ Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Mass

Value for mass

Center_x

x coordinate for center of gravity

Center_y

y coordinate for center of gravity

Center_z

z coordinate for center of gravity

Config_opt

Configuration options as defined in [swInConfigurationOpts_e](#)

Config_names

Array of the configuration names

Return Value

True if the mass and center of gravity are set, false if not

 Visual Basic for Applications (VBA) Syntax

See [MassProperty::SetAssignedMassProp](#).

 Remarks

If...	Then...
You are editing a subcomponent	you are setting the mass for this subcomponent and not for the top-level model.
You specify a value < 0 for Mass	mass is calculated; it is not user-defined.
Config_opt is set to swSpecifyConfiguration	Config_names is used.

 See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::ISetAssignedMassProp Method](#)
[IMassProperty::Mass Property](#)
[IMassProperty::UserAssigned Property](#)
[IMassProperty::IGetCenterOfMass Method](#)
[IMassProperty::CenterOfMass Property](#)

 Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

Sets the coordinate system to use when calculating mass properties for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function SetCoordinateSystem( _  
    ByVal Coords As MathTransform _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IMassProperty  
Dim Coords As MathTransform  
Dim value As System.Boolean  
  
value = instance.SetCoordinateSystem(Coords)
```

C#

```
System.bool SetCoordinateSystem(  
    MathTransform Coords  
)
```

C++/CLI

```
System.bool SetCoordinateSystem(  
& MathTransform^ Coords  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Coords
[Math transform](#)

Return Value

True if the coordinate system is set, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty::SetCoordinateSystem](#).

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IModelDoc2::GetCurrentCoordinateSystemName Method](#)
[IModelDoc2::InsertCoordinateSystem Method](#)
[IModelDocExtension::GetCoordinateSystemTransformByName Method](#)

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetOverrideCenterOfMassValue Method (IMassProperty)

[See Also](#) [Example](#)



SolidWorks.Interop.sldworks Namespace > IMassProperty Interface : SetOverrideCenterOfMassValue Method (IMassProperty)

Overrides the center of mass of the model currently being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverrideCenterOfMassValue( _
    ByVal Value As System.Object, _
    ByVal CoordinateSystemName As System.String, _
    ByVal Config\_option As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim Value As System.Object
Dim CoordinateSystemName As System.String
Dim Config_option As System.Integer
Dim Config_names As System.Object
Dim value As System.Boolean

value = instance.SetOverrideCenterOfMassValue(Value, CoordinateSystemName, Config_option, Config_names)
```

C#

```
System.bool SetOverrideCenterOfMassValue(
    System.object Value,
    System.string CoordinateSystemName,
    System.int Config\_option,
    System.object Config\_names
)
```

C++/CLI

```
System.bool SetOverrideCenterOfMassValue(
    & System.Object^ Value,
    & System.String^ CoordinateSystemName,
    & System.int Config\_option,
    & System.Object^ Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value

Array of three doubles of the x, y, and z coordinates of the center of mass

CoordinateSystemName

Name of the coordinate system in which the center of mass is defined

Config_option

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of configuration names; valid only if Config_option = swInConfigurationOpts_e.swSpecifyConfiguration

Return Value

True if the center of mass is overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty::SetOverrideCenterOfMassValue](#).

Example

[Get and Override Mass Properties \(VBA\)](#)

[Get and Override Mass Properties \(VB.NET\)](#)

[Get and Override Mass Properties \(C#\)](#)

See Also

[IMassProperty Interface](#)

[IMassProperty Members](#)

[IMassProperty::ISetOverrideCenterOfMassValue Method](#)

[IMassProperty::OverrideCenterOfMass Property](#)

Availability

SetOverrideMassValue Method (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : SetOverrideMassValue Method (IMassProperty)

Overrides the mass of the model currently being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverrideMassValue( _
    ByVal Value As System.Double, _
    ByVal Config\_option As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim Value As System.Double
Dim Config_option As System.Integer
Dim Config_names As System.Object
Dim value As System.Boolean

value = instance.SetOverrideMassValue(Value, Config_option, Config_names)
```

C#

```
System.bool SetOverrideMassValue(
    System.double Value,
    System.int Config\_option,
    System.object Config\_names
)
```

C++/CLI

```
System.bool SetOverrideMassValue(
    & System.double Value,
    & System.int Config\_option,
    & System.Object^ Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value

Override mass value (see **Remarks**)

Config_option

Configuration option as defined in [swInConfigurationOptions_e](#)

Config_names

Array of configuration names (see **Remarks**)

Return Value

True if the mass value is overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty::SetOverrideMassValue](#).

Example

[Get and Override Mass Properties \(VBA\)](#)

[Get and Override Mass Properties \(VB.NET\)](#)

[Get and Override Mass Properties \(C#\)](#)

Remarks

If...	Then...
You are editing a subcomponent	you are overriding the mass for this subcomponent and not for the top-level model.
Value > 0	mass is overridden.
Value < 0	mass is calculated.
Config_option = swInConfigurationOpts_e.swSpecifyConfiguration	Config_names is used.

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::!SetOverrideMassValue Method](#)
[IMassProperty::OverrideMass Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetOverrideMomentsOfInertiaValue Method (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : SetOverrideMomentsOfInertiaValue Method (IMassProperty)

Overrides the moments of inertia of the model currently being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverrideMomentsOfInertiaValue( _
    ByVal ReferenceFrame As System.Integer, _
    ByVal CoordinateSystemName As System.String, _
    ByVal Value As System.Object, _
    ByVal Config\_option As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim ReferenceFrame As System.Integer
Dim CoordinateSystemName As System.String
Dim Value As System.Object
Dim Config_option As System.Integer
Dim Config_names As System.Object
Dim value As System.Boolean
```

```
value = instance.SetOverrideMomentsOfInertiaValue(ReferenceFrame, CoordinateSystemName, Value,
Config_option, Config_names)
```

C#

```
System.bool SetOverrideMomentsOfInertiaValue(
    System.int ReferenceFrame,
    System.string CoordinateSystemName,
    System.object Value,
    System.int Config\_option,
    System.object Config\_names
)
```

C++/CLI

```

System.bool SetOverrideMomentsOfInertiaValue(
& System.int ReferenceFrame,
& System.String^ CoordinateSystemName,
& System.Object^ Value,
& System.int Config\_option,
& System.Object^ Config\_names
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ReferenceFrame

Frame of reference as defined in [swMomentsOfInertiaReferenceFrame_e](#)

CoordinateSystemName

Name of coordinate system; valid only if ReferenceFrame = swMomentsOfInertiaReferenceFrame_e.swMomentsOfInertiaReferenceFrame_UserCoordinateSystem

Value

Array of nine doubles: [Lxx, Lxy, Lxz, Lyx, Lyy, Lyz, Lzx, Lzy, Lzz]

Config_option

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of configuration names; valid only if Config_option = swInConfigurationOpts_e.swSpecifyConfiguration

Return Value

True if the moments of inertia are overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty::SetOverrideMomentsOfInertiaValue](#).

Example

[Get and Override Mass Properties \(VBA\)](#)

[Get and Override Mass Properties \(VB.NET\)](#)

[Get and Override Mass Properties \(C#\)](#)

See Also

[IMassProperty Interface](#)

[IMassProperty Members](#)

[IMassProperty::SetOverrideMomentsOfInertiaValue Method](#)

[IMassProperty::OverrideMomentsOfInertia Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetOverridePrincipleAxesOrientation Method (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : SetOverridePrincipleAxesOrientation Method (IMassProperty)

Overrides the orientation of the specified principal axis of inertia for the model currently being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```

Function SetOverridePrincipleAxesOrientation( _
    ByVal Axis As System.Integer, _
    ByVal Value As System.Object, _
    ByVal Config\_option As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Boolean

```

Visual Basic (Usage)

```

Dim instance As IMassProperty
Dim Axis As System.Integer
Dim Value As System.Object
Dim Config_option As System.Integer

```



```
Dim Config_names As System.Object
Dim value As System.Boolean
```

```
value = instance.SetOverridePrincipleAxesOrientation(Axis, Value, Config_option, Config_names)
```

C#

```
System.bool SetOverridePrincipleAxesOrientation(
    System.int Axis,
    System.object Value,
    System.int Config\_option,
    System.object Config\_names
)
```

C++/CLI

```
System.bool SetOverridePrincipleAxesOrientation(
    & System.int Axis,
    & System.Object^ Value,
    & System.int Config\_option,
    & System.Object^ Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Axis

One of the following principal axes:

- 0 = X axis
- 1 = Y axis
- 2 = Z axis

Value

An array of three doubles of the x, y, and z coordinates of Axis

Config_option

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of configuration names; valid only if Config_option = swInConfigurationOpts_e.swSpecifyConfiguration

Return Value

True if orientation of principal axis is overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty::SetOverridePrincipleAxesOrientation](#).

See Also

[IMassProperty Interface](#)

[IMassProperty Members](#)

[IMassProperty::SetOverridePrincipleAxesOrientation Method](#)

[IMassProperty::PrincipleAxesOfInertia Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetOverridePrincipleMomentsOfInertia Method (IMassProperty)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : SetOverridePrincipleMomentsOfInertia Method (IMassProperty)

Overrides the principal moments of inertia of the model currently being edited in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverridePrincipleMomentsOfInertia( _
    ByVal Value As System.Object, _
    ByVal Config\_option As System.Integer, _
```

```
ByVal Config\_names As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim Value As System.Object  
Dim Config_option As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Boolean  
  
value = instance.SetOverridePrincipleMomentsOfInertia(Value, Config_option, Config_names)
```

C#

```
System.bool SetOverridePrincipleMomentsOfInertia(  
    System.object Value,  
    System.int Config\_option,  
    System.object Config\_names  
)
```

C++/CLI

```
System.bool SetOverridePrincipleMomentsOfInertia(  
& System.Object^ Value,  
& System.int Config\_option,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value

Array of three doubles of the principal moments of inertia: [Px, Py, Pz]

Config_option

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of configuration names; valid only if Config_option = swInConfigurationOpts_e.swSpecifyConfiguration

Return Value

True if the principal moments of inertia are overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty::SetOverridePrincipleMomentsOfInertia](#).

Example

[Get and Override Mass Properties \(VBA\)](#)

[Get and Override Mass Properties \(VB.NET\)](#)

[Get and Override Mass Properties \(C#\)](#)

See Also

[IMassProperty Interface](#)

[IMassProperty Members](#)

[IMassProperty::SetOverridePrincipleMomentsOfInertia Method](#)

[IMassProperty::OverrideMomentsOfInertia Property](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SurfaceArea Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : SurfaceArea Property (IMassProperty)

Gets the surface area for this model.

Visual Basic Syntax

Visual Basic (Declaration)

```
ReadOnly Property SurfaceArea As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim value As System.Double

value = instance.SurfaceArea
```

C#

```
System.double SurfaceArea {get;}
```

C++/CLI

```
property System.double SurfaceArea {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Surface area

Visual Basic for Applications (VBA) Syntax

See [MassProperty::SurfaceArea](#).

Example

[Insert Coordinate System at Center of Mass \(VBA\)](#)
[Set Dimensions to Mid-tolerance \(VBA\)](#)
[Get and Override Mass Properties \(C#\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(VBA\)](#)
[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

UserAssigned Property (IMassProperty)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : UserAssigned Property (IMassProperty)

Gets whether the mass property values are user-defined or calculated for this document, regardless of which model is being edited.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property UserAssigned As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim value As System.Boolean
```

```
instance.UserAssigned = value
value = instance.UserAssigned
```

C#

```
System.bool UserAssigned {get; set;}
```

C++/CLI

```
property System.bool UserAssigned {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if mass property values are user-defined, false if calculated

Visual Basic for Applications (VBA) Syntax

See [MassProperty::UserAssigned](#).

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)
[IMassProperty::ISetAssignedMassProp Method](#)
[IMassProperty::SetAssignedMassProp Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UseSystemUnits Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : UseSystemUnits Property (IMassProperty)

Gets and sets whether to use system units or document units when calculating mass properties for this model.

.NET Syntax

Visual Basic (Declaration)

```
Property UseSystemUnits As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty
Dim value As System.Boolean
```

```
instance.UseSystemUnits = value
value = instance.UseSystemUnits
```

C#

```
System.bool UseSystemUnits {get; set;}
```

C++/CLI

```
property System.bool UseSystemUnits {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use system units, false to use document units

Visual Basic for Applications (VBA) Syntax

See [MassProperty::UseSystemUnits](#).

Example

[Get and Override Mass Properties \(VBA\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(C#\)](#)

Remarks

The default value is True. Thus, system units (meters, radians, and kilograms) are used. All properties returning a value are adjusted accordingly

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

Volume Property (IMassProperty)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty Interface](#) : Volume Property (IMassProperty)

Gets the volume of this model.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Volume As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMassProperty  
Dim value As System.Double
```

```
value = instance.Volume
```

C#

```
System.double Volume {get;}
```

C++/CLI

```
property System.double Volume {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Volume

Visual Basic for Applications (VBA) Syntax

See [MassProperty::Volume](#).

Example

[Insert Coordinate System at Center of Mass \(VBA\)](#)
[Set Dimensions to Mid-Tolerance \(VBA\)](#)
[Get and Override Mass Properties \(C#\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(VBA\)](#)
[Get Mass Properties of Multibody Assembly Component \(C#\)](#)
[Get Mass Properties of Multibody Assembly Component \(VB.NET\)](#)
[Get Mass Properties of Multibody Assembly Component \(VBA\)](#)

See Also

[IMassProperty Interface](#)
[IMassProperty Members](#)

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2